Saikarthik Ramakrishnan

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Education

Shiv Nadar Institution of Eminence (SNIoE) Delhi NCR, India
B.Tech., Electrical & Computer Engineering *2025 – 2029*
CGPA **8.55**/10; **Dean's List**, Monsoon 2025 (SGPA 8.85)

BVM Global, Perungudi (CBSE): *Class XII: 93%, School Topper in Artificial Intelligence (100%)* Chennai

Vaels International School (ICSE): *Class X: 96%, 3rd in school (Science)* Chennai

Experience

Engineering Intern Jul 2026 – Present
SPowerZ Solutions, Camera-Based Passenger Counting System (CPCS) *India*

- **Edge vision pipeline:** Developing a **bus passenger counter** using **YOLOv8** detection and **ByteTrack** tracking on an **Orange Pi 5**, logging trips to SQLite with a bilingual (English/Hindi) dashboard.
- **Robustness and cost:** Improving counts under **crowding and occlusion**, and benchmarking **accuracy against hardware cost** across camera resolutions and frame rates.

Projects

Parallel Cellular Automaton Engine on FPGA:
Hardware engine that gives **every grid cell its own logic circuit** so the whole grid updates at once. Runs the Drossel–Schwabl **forest-fire model** on a Gowin GW2A-18 FPGA, streams frames over **UART** to a live browser console, verified with **cocotb**, and uses about **5× fewer LUTs** after synthesis tuning. *(Verilog, cocotb, Python)*

Self-Healing Reconfigurable Analog Circuit:
ESP32-controlled RC filter that tunes itself to a requested cutoff frequency and **reroutes around failing components**. A **genetic algorithm** guided by a **Gaussian-process ensemble** searches **2,040 circuit paths** and keeps pre-tested backup routes; across 90 simulated fault runs it **recovered from every fault** with a median of 3 probes. *(Python, NumPy, ESP32, PlatformIO)*

Evolutionary Game Theory for Self-Driving Traffic:
Simulation testing whether **autonomous and human drivers learn to cooperate**, with driving strategies spreading through the fleet by payoff. Found that a **3% subsidy** for verified autonomous-vehicle firmware **removes the cooperation dilemma** where tailgating taxes and headway rules fail. Submitted to **Hack The Limit**. *(JavaScript)*

Agent-Based Traffic Jam Simulator:
Shows how **jams form "out of nowhere"** on a ring road, using **IDM car-following** and a corrected **FollowerStopper** autonomous-car controller, **validated against the Sugiyama (2008)** jam-wave experiment. *(JavaScript)*

Research-Paper Question Answering with RAG:
Answers questions about uploaded papers with **citations to the exact passages**. Combines **FAISS** dense search and **BM25** with rank fusion and a **cross-encoder reranker** behind FastAPI and Streamlit; fixed a silent chunking bug that was **discarding about 60%** of each passage. *(Python, FastAPI, Streamlit)*

Spiking Neural Network Accelerator for Gesture Decoding: Ongoing
Event-driven FPGA accelerator that decodes **hand gestures from forearm muscle (EMG) signals** for **low-power prosthetic control**, computing only when a neuron spikes. *(Verilog, snnTorch)*

Research & Achievements

- **Battery health estimation audit** *(manuscript in preparation, IEEE Access):* showed on NASA PCoE and Oxford data that **random-split testing overstates accuracy about 4×** versus leave-one-cell-out testing.
- **School Topper**, CBSE Class XII Artificial Intelligence (100%). **Coursera:** Generative AI with Developers (Specialization), Battery Technologies, RAG.

Technical Skills

Languages:	Python, C, C++, Java, JavaScript, Verilog, MATLAB, HTML/CSS
Machine Learning:	PyTorch, TensorFlow, scikit-learn, snnTorch, pandas, NumPy, Gaussian processes, genetic algorithms, SNNs
Computer Vision:	YOLOv8, ByteTrack, OpenCV, object detection, multi-object tracking, edge inference
LLMs & Web:	RAG, FAISS, BM25, sentence-transformers, rerankers, React, FastAPI, Flask, Streamlit, SQLite
Hardware:	FPGA (Gowin, Yosys, nextpnr), cocotb, Orange Pi 5, ESP32, PlatformIO, Arduino, UART, MQTT
Tools:	Git, GitHub, Linux, Cython, LaTeX